
EHRMAN CREST CIRC | GRADES 5 AND 6

CIRC Teacher Guide

All 31 remaining contacts. 35 minutes each. Grade 5 CIRC Tank Jr and Grade 6 CIRC Tank supports on every page.

One contact, one two-page spread

A contact is one class meeting per class, not one calendar day.

Every contact gets the same two pages. The first page is the one you keep open while the class runs: materials, the one-minute demo, the timed steps, and the exit check. The second page is the one you read before the class: grade supports, cleanup, fast finish, safety, and the conditions that stay visible. Contacts 22 and 28 add a third page holding a complete no-equipment fallback lesson.

Contacts 1 to 21 are the prepared CIRC lessons from the master route. Contacts 22 to 31 reuse existing catalog projects, and their full timed directions are printed here so you never have to go looking for a linked lesson mid-class. All 36 catalog alternatives stay available; this route uses ten of them and replaces none.

Open the next unused contact for that class. Show the goal, run the one-minute demo, then let students work, check, and improve. Keep the timer teacher-controlled. Save unfinished work in the class tray. A spoken explanation or a demonstration counts as evidence.

Classes move at different speeds, so track each class by contact number rather than by date. Closures, assemblies, and absences shift the calendar; the contact order does not change. Keep any extra contact as a buffer.

GRADE SUPPORTS ON CONTACTS 22 TO 31

The catalog states one core plan for grades 5 and 6. The grade rows on those pages apply the route's own stated expectations, Tank Jr bounded and supported, Tank measurable and justified, to each lesson's own evidence. They add support, not new content.

OPTIONAL AND TEACHER-ONLY EQUIPMENT

- ChompSaw: optional. Scissors and teacher-prepared pieces complete contact 4. The machine station opens only after separate school approval and its exact demonstration.
- Resin: contact 6 is dry planning only. Students never mix, pour, heat, or demold. Only a small teacher trial is being considered, pending product and school procedure.
- Scanner or digitizer: hardware is not identified. Demonstrate the actual approved device. The paper preview is a complete fallback.
- Hot glue and knife cuts: teacher station only. Tape and tabs complete the build.
- Thermometers, scales, calculators: optional. Every lesson works without them.
- Tubing cutter on contact 26: teacher-operated. Students mark and hand over.
- Micro:bits on contact 28: existing approved devices, or the complete paper-code version printed on its own page. No purchase is assumed.
- Vacuum former and graphics printer: not needed. Cardboard packaging is complete.

NOT ASSUMED ANYWHERE IN THIS GUIDE

- No dates. Contacts are class meetings and are tracked per class.
- No purchases, new accounts, or new online assignments.
- No student resin casting, class-set molds, or bulk rollout.
- No outside visitors and no approved showcase event.
- No student names, private schedule details, or personal records.

THE ROUTE

All 31 contacts in order

Same order for every class. Mark a class complete by contact, not by date.

Contact	Lesson	Source	Purpose
1	Neuro art and book checkout	Core route	I can join curved lines, round their corners, and follow our book-checkout routine.
2	Turn artwork into a digital image	Core route	I can capture a clear image, name it clearly, and check that another person can find it.
3	Design a logo people can read	Core route	I can combine a simple symbol and readable lettering, then improve them after a quick test.
4	Cardboard joins and careful cutting	Core route	I can make a stable cardboard connection and test it before adding decoration.
5	Make a package that protects	Core route	I can protect a model product, test the package, and improve one weak point.
6	Plan a small casting without pouring	Core route	I can draw a useful small object, estimate its size, and compare possible materials.
7	Tank 1: Notice a useful problem	Core route	I can notice a real, manageable problem and explain who it affects.
8	Tank 2: Listen before choosing	Core route	I can ask a useful question and use the answer to improve my problem statement.
9	Tank 3: Try three ideas	Core route	I can sketch different solutions and choose one using a clear reason.
10	Tank 4: Plan the first build	Core route	I can make a labeled plan and choose a small, testable first version.
11	Tank 5: Practice the tricky part	Core route	I can practice one connection or moving part before using it in my prototype.
12	Tank 6: Build version one	Core route	I can build a rough model that shows my idea's main job.
13	Tank 7: Plan a fair test	Core route	I can plan a repeatable test that checks what my design claims.
14	Tank 8: Test and record	Core route	I can run the same test three times and record what actually happens.
15	Tank 9: Improve one feature	Core route	I can use test evidence to change one feature and check whether it helps.
16	Tank 10: Let someone try	Core route	I can watch a user try my idea and learn without coaching every move.
17	Tank 11: Use materials wisely	Core route	I can choose materials that do the job while reducing unnecessary waste.
18	Tank 12: Name and package	Core route	I can communicate my product's purpose with a clear name, logo, and simple package.
19	Tank 13: Explain your idea	Core route	I can explain the problem, my solution, and evidence that helped me improve it.
20	Tank 14: Rehearse and refine	Core route	I can practice a clear presentation and improve it after specific feedback.
21	Tank 15: Share and reflect	Core route	I can show my design, listen to feedback, and explain how it changed.
22	Outdoor Microclimate Map	Catalog 6	Observe and compare outdoor places. Thermometers are optional.

Contact	Lesson	Source	Purpose
23	Seed Travelers	Catalog 7	Test a seed-inspired carrier with shared test stations.
24	Paper Bridge	Catalog 5	Compare a small structure using the same reusable loads.
25	Moving Picture Machine	Catalog 18	Make and revise a short paper animation.
26	Hydroponics Flow Lab	Catalog 26	Test a gravity-fed water model, not a purchased growing installation.
27	Stormwater Rescue	Catalog 27	Use shared trays to compare runoff before and after one change.
28	Micro:bit Sensor Station	Catalog 28	Use existing approved devices or the complete paper-code fallback.
29	Cardboard Arcade	Catalog 10	Combine the year's connection and testing skills in a compact game.
30	Tech Terrarium	Catalog 2	Make a teacher-approved shared contribution while preserving the existing habitat.
31	Reflect, reset, and catch up	Catalog 36	Use Demo Day reflection and reset, or complete a missed core contact.

SHARED SUPPLY PLAN

- Core: school paper, pencils, rulers, scissors, markers, existing cardboard, tape, trays, craft sticks, string, fasteners, and reusable test objects.
- Neuro art uses the existing art plan. The checkout lesson stays dry.
- Scan and logo work use existing approved devices. Paper planning is the device-free fallback.
- Spring flow and runoff share six-group trays, cups, tubing, towels, and model materials.
- Use existing micro:bits if available. The paper-code fallback needs no electronics.
- Showcase work reuses prototypes, labels, class surfaces, and test records.
- Subtract usable school stock before purchasing. This guide is not an inventory or an order.

Neuro art and book checkout

I can join curved lines, round their corners, and follow our book-checkout routine.

Core route lesson circ-neuro

MATERIALS

Watercolor paper ■ Black drawing markers ■ Washable color markers ■ Class artwork trays ■ Existing library books and checkout system

DEMO IN ONE MINUTE

- Watch four curved lines cross.
- Turn one sharp corner into a smooth curve.
- Notice the large spaces left for color.

Min	Step	What students actually do	Teacher move
3	See today's plan	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can join curved lines, round their corners, and follow our book-checkout routine.
4	Watch the line	Watch four curved lines cross. Turn one sharp corner into a smooth curve. Notice the large spaces left for color.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
13	Draw and round	Write your name and class on the back. Draw four to six crossing curves. Round the sharp corners. Add dry color if time allows.	Pause each group early. Ask what its next useful action will be. Use this short, dry-art version with checkout. Save wet blending for another contact using the existing neuro guide.
10	Choose your book	Cap markers and place artwork in the class tray. When called, follow our library checkout routine. Return with your book and read quietly.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Point to one corner you rounded and show that your table is ready.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK **Point to one corner you rounded and show that your table is ready.**

NOTES FOR NEXT TIME

6

Turn artwork into a digital image

I can capture a clear image, name it clearly, and check that another person can find it.

Core route lesson circ-digitize

MATERIALS

Existing school-approved camera or scanner ■ Existing school device and approved storage ■ Artwork ■ Paper ■ Pencils

DEMO IN ONE MINUTE

- Watch a page being placed flat in good light.
- Compare a crooked image with a cropped, readable image.
- Watch the teacher save one example.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can capture a clear image, name it clearly, and check that another person can find it.
5	Watch and try	Watch a page being placed flat in good light. Compare a crooked image with a cropped, readable image. Watch the teacher save one example.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Use the school-approved camera or scanner shown by your teacher. Capture only your artwork. Keep faces and private information outside the image. Save using the class code and project label shown on the board.	Pause each group early. Ask what its next useful action will be. Hardware model is unknown. Demonstrate the actual device; do not invent its buttons. Paper planning completes the fallback.
5	Check one thing	Find your saved image again. Check the edges, direction, and readability. If no device is available, draw a paper preview and write its file label.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Show a readable saved image, or a paper preview with a clear file label.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK

Show a readable saved image, or a paper preview with a clear file label.

NOTES FOR NEXT TIME

3

Design a logo people can read

I can combine a simple symbol and readable lettering, then improve them after a quick test.

Core route lesson circ-logo

MATERIALS

Paper ■ Pencils ■ Erasers ■ Rulers ■ Shared markers ■ Existing approved digital image tool if available

DEMO IN ONE MINUTE

- Compare a clear symbol with a crowded symbol.
- Look at the same logo large and small.
- Notice how light and dark areas help the words stand out.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can combine a simple symbol and readable lettering, then improve them after a quick test.
5	Watch and try	Compare a clear symbol with a crowded symbol. Look at the same logo large and small. Notice how light and dark areas help the words stand out.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Choose a class project or invented product. Sketch three small logo ideas. Choose one. Use a simple symbol and no more than three colors.	Pause each group early. Ask what its next useful action will be. Paper is the complete default. Digital cleanup is optional and does not require a new account.
5	Check one thing	Show the logo without explaining it. Ask what the viewer notices and reads first. Change one unclear detail and test again.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Show your logo and name the change that made it clearer.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK **Show your logo and name the change that made it clearer.**

NOTES FOR NEXT TIME

GRADE SUPPORTS

Grade 6 CIRC Tank Name the audience. Explain how the symbol, lettering, and contrast suit that audience.

Fast finish. Improve the same work or rehearse your explanation. Ask before taking more materials.

- Paper is the complete default. Digital cleanup is optional and needs no new account.
- Teacher-led. No new student account, grade, or online assignment.
- Keep student names off public screens. Use class storage codes.

Next: 4. Cardboard joins and careful cutting

Cardboard joins and careful cutting

I can make a stable cardboard connection and test it before adding decoration.

Core route lesson circ-cardboard

MATERIALS

Clean cardboard ■ Paper ■ Pencils ■ Rulers ■ Student scissors ■ Masking tape ■ Reusable trays ■ Teacher-prepared pieces if needed ■ Optional approved ChompSaw station

DEMO IN ONE MINUTE

- Watch the teacher cut away from the holding hand.
- Compare a tab, slot, and taped join.
- Watch the teacher test each join gently.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can make a stable cardboard connection and test it before adding decoration.
5	Watch and try	Watch the teacher cut away from the holding hand. Compare a tab, slot, and taped join. Watch the teacher test each join gently.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Sketch a small freestanding shape. Choose a tab, slot, or taped join. Cut with approved scissors or use teacher-prepared pieces. Assemble your shape.	Pause each group early. Ask what its next useful action will be. ChompSaw purchase and operating approval are not assumed. Follow its manual and supervision rules if supplied. Teacher handles knife cuts and hot glue.
5	Check one thing	Place your shape on the table and test its balance. Change one weak join. Compare the first and improved versions.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Show one stable connection and explain why it holds.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK **Show one stable connection and explain why it holds.**

NOTES FOR NEXT TIME

Grade supports, reset, and the conditions that stay visible.

GRADE SUPPORTS

Grade 5 CIRC Tank Jr Build one stable shape using a demonstrated connection.

Grade 6 CIRC Tank Compare two connection types and explain the tradeoff before choosing.

CLEANUP, EXIT, AND EARLY FINISHERS

Cleanup. Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Exit check. Show one stable connection and explain why it holds.

Fast finish. Improve the same work or rehearse your explanation. Ask before taking more materials.

SAFETY AND CONDITIONS

Scissors are the default. A ChompSaw station opens only after school approval and the exact machine demonstration.

- Scissors and teacher-prepared pieces complete this lesson.
- A ChompSaw station opens only after separate school approval and its exact operating demonstration.
- Teacher-led. No new student account, grade, or online assignment.
- Keep student names off public screens. Use class storage codes.

Previous: 3. Design a logo people can read

Next: 5. Make a package that protects

NOTES FOR NEXT TIME

Make a package that protects

I can protect a model product, test the package, and improve one weak point.

Core route lesson circ-packaging

MATERIALS

Reclaimed small boxes ■ Clean cardboard ■ Paper ■ String ■ Masking tape ■ Model products ■ Rulers ■ Scissors ■ Shared scale if available

DEMO IN ONE MINUTE

- Watch a model product move inside a loose package.
- Notice padding, support, and an opening.
- Watch one teacher-approved handling test.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can protect a model product, test the package, and improve one weak point.
5	Watch and try	Watch a model product move inside a loose package. Notice padding, support, and an opening. Watch one teacher-approved handling test.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Choose the reusable model product provided. Sketch a package that opens without damage. Build using a material limit shown by your teacher.	Pause each group early. Ask what its next useful action will be. Reuse catalog lesson 24, Package Rescue. A vacuum former is not needed for this complete lesson.
5	Check one thing	Run the same gentle handling test twice. Check movement, damage, and ease of opening. Improve one weak point without adding unnecessary material.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Demonstrate how your package protects and opens, and explain one improvement.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK

Demonstrate how your package protects and opens, and explain one improvement.

NOTES FOR NEXT TIME

Grade supports, reset, and the conditions that stay visible.

GRADE SUPPORTS

Grade 5 CIRC Tank Jr Protect one simple object and show one before-and-after change.

Grade 6 CIRC Tank Compare protection, material use, and the user's opening experience.

CLEANUP, EXIT, AND EARLY FINISHERS

Cleanup. Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Exit check. Demonstrate how your package protects and opens, and explain one improvement.

Fast finish. Improve the same work or rehearse your explanation. Ask before taking more materials.

SAFETY AND CONDITIONS

Only use the teacher's handling test. Do not throw packages or test near people or equipment.

- Cardboard packaging is complete without a vacuum former or graphics printer.
- Teacher-led. No new student account, grade, or online assignment.
- Keep student names off public screens. Use class storage codes.

Previous: 4. Cardboard joins and careful cutting

Next: 6. Plan a small casting without pouring

NOTES FOR NEXT TIME

Plan a small casting without pouring

I can draw a useful small object, estimate its size, and compare possible materials.

Core route lesson circ-resin-plan

MATERIALS

Paper ■ Pencils ■ Rulers ■ Calculators if available ■ Empty dry mold or dimensioned drawing ■ Teacher-cleared cured example or picture

DEMO IN ONE MINUTE

- Look at a dry mold or its measured drawing.
- Identify the inside space that shapes the object.
- Compare a cardboard model with a teacher-cleared cured example or picture.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can draw a useful small object, estimate its size, and compare possible materials.
5	Watch and try	Look at a dry mold or its measured drawing. Identify the inside space that shapes the object. Compare a cardboard model with a teacher-cleared cured example or picture.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Sketch a small token, tag, or game piece. Measure a rectangular practice cavity or its drawing. Use length, width, and depth to estimate its volume.	Pause each group early. Ask what its next useful action will be. Only a small teacher trial is planned, pending the actual product, school procedure, and Home Depot. No student casting rollout is approved.
5	Check one thing	Compare making your design from cardboard and casting material. Name one reason for each choice. Improve the design using the teacher's size limit.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Show a labeled design and explain why your chosen material fits its use.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK **Show a labeled design and explain why your chosen material fits its use.**

NOTES FOR NEXT TIME

GRADE SUPPORTS

Grade 6 CIRC Tank	Calculate the rectangular practice volume and explain why an irregular mold needs a different measurement method.
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Fast finish. Improve the same work or rehearse your explanation. Ask before taking more materials.

- Students do not mix, pour, heat, demold, or handle uncured resin.
- Only a small teacher trial is being considered, pending the selected product and the school procedure. No student casting is approved.
- Teacher-led. No new student account, grade, or online assignment.
- Keep student names off public screens. Use class storage codes.

Next: 7. Tank 1: Notice a useful problem

Tank 1: Notice a useful problem

I can notice a real, manageable problem and explain who it affects.

Core route lesson tank-notice

MATERIALS

Paper ■ Pencils ■ Teacher-approved problem pictures

DEMO IN ONE MINUTE

- Compare a broad wish with one specific everyday problem.
- Notice who is affected and when it happens.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can notice a real, manageable problem and explain who it affects.
5	Watch and try	Compare a broad wish with one specific everyday problem. Notice who is affected and when it happens.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Look at the approved classroom examples. List or sketch three small problems. Choose one that can be explored with classroom materials.	Pause each group early. Ask what its next useful action will be. Do not identify classmates as problems. Model a practical task or object that could work better.
5	Check one thing	Tell a partner who has the problem. Explain when it happens. Narrow any idea that needs money, machinery, or changes outside our control.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Complete: Someone needs help with this problem when this happens.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK

Complete: Someone needs help with this problem when this happens.

NOTES FOR NEXT TIME

GRADE SUPPORTS

Grade 6 CIRC Tank Find and describe a specific user problem with one direct observation.

Fast finish. Improve the same work or rehearse your explanation. Ask before taking more materials.

- Do not name a classmate as the problem. Model a task or object that could work better.
- Teacher-led. No new student account, grade, or online assignment.
- Keep student names off public screens. Use class storage codes.

Next: 8. Tank 2: Listen before choosing

Tank 2: Listen before choosing

I can ask a useful question and use the answer to improve my problem statement.

Core route lesson tank-listen

MATERIALS

Paper ■ Pencils ■ Reusable user cards

DEMO IN ONE MINUTE

- Compare a leading question with an open question.
- Watch a short respectful interview.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can ask a useful question and use the answer to improve my problem statement.
5	Watch and try	Compare a leading question with an open question. Watch a short respectful interview.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Prepare two questions about the task or object. Ask an approved partner, or use the teacher's user card. Record the useful answer with a short note or drawing.	Pause each group early. Ask what its next useful action will be. Use in-class role-play or approved users. Do not contact outside people or record private information.
5	Check one thing	Compare the answer with your first idea. Change the problem statement if needed. Ask whether you are describing a need or jumping to a solution.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Share one thing you learned from the user and what it changed.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK **Share one thing you learned from the user and what it changed.**

NOTES FOR NEXT TIME

GRADE SUPPORTS

Grade 6 CIRC Tank Ask an open follow-up and separate what was observed from what was assumed.

Fast finish. Improve the same work or rehearse your explanation. Ask before taking more materials.

- In-class role play or approved users only. No outside contact and no private information.
- Teacher-led. No new student account, grade, or online assignment.
- Keep student names off public screens. Use class storage codes.

Next: 9. Tank 3: Try three ideas

Tank 3: Try three ideas

I can sketch different solutions and choose one using a clear reason.

Core route lesson tank-options

MATERIALS

Paper ■ Pencils ■ Rulers ■ Reusable criteria cards

DEMO IN ONE MINUTE

- See three different ways to solve one small problem.
- Compare usefulness, simplicity, and available materials.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can sketch different solutions and choose one using a clear reason.
5	Watch and try	See three different ways to solve one small problem. Compare usefulness, simplicity, and available materials.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Sketch three different solutions. Label the part that does the work. Circle one idea you can test with classroom materials.	Pause each group early. Ask what its next useful action will be.
5	Check one thing	Show the ideas to a partner. Name one strength and one concern for the chosen idea. Revise the sketch before building.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Show three ideas and explain why one is the best first test.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK **Show three ideas and explain why one is the best first test.**

NOTES FOR NEXT TIME

Grade supports, reset, and the conditions that stay visible.

GRADE SUPPORTS

Grade 5 CIRC Tank Jr Draw two ideas independently, then create a third with a prompt or partner.

Grade 6 CIRC Tank Use two chosen criteria to compare all three ideas.

CLEANUP, EXIT, AND EARLY FINISHERS

Cleanup. Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Exit check. Show three ideas and explain why one is the best first test.

Fast finish. Improve the same work or rehearse your explanation. Ask before taking more materials.

SAFETY AND CONDITIONS

Use teacher-approved materials and tools. Keep paths clear. Return tools before moving.

- Teacher-led. No new student account, grade, or online assignment.
- Keep student names off public screens. Use class storage codes.

Previous: 8. Tank 2: Listen before choosing

Next: 10. Tank 4: Plan the first build

NOTES FOR NEXT TIME

Tank 4: Plan the first build

I can make a labeled plan and choose a small, testable first version.

Core route lesson tank-plan

MATERIALS

Clean cardboard ■ Paper ■ Pencils ■ Rulers ■ Student scissors ■ Masking tape ■ Reusable trays

DEMO IN ONE MINUTE

- Watch a sketch become a parts list.
- Separate essential working parts from decoration.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can make a labeled plan and choose a small, testable first version.
5	Watch and try	Watch a sketch become a parts list. Separate essential working parts from decoration.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Draw the first version from a useful view. Label its working parts and approximate sizes. Choose materials and agree on the first build jobs.	Pause each group early. Ask what its next useful action will be. Limit each team to a small tray of materials. No shopping or new equipment is needed for the first prototype.
5	Check one thing	Explain how a user will try the model. Remove any part that does not help the first test. Show the teacher your plan before collecting supplies.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Show a labeled plan, short materials list, and one thing your model must do.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK **Show a labeled plan, short materials list, and one thing your model must do.**

NOTES FOR NEXT TIME

Grade supports, reset, and the conditions that stay visible.

GRADE SUPPORTS

Grade 5 CIRC Tank Jr Use a three-box plan: problem, sketch, and first test.

Grade 6 CIRC Tank Include a material limit, approximate dimensions, and a measurable success criterion.

CLEANUP, EXIT, AND EARLY FINISHERS

Cleanup. Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Exit check. Show a labeled plan, short materials list, and one thing your model must do.

Fast finish. Improve the same work or rehearse your explanation. Ask before taking more materials.

SAFETY AND CONDITIONS

Use teacher-approved materials and tools. Keep paths clear. Return tools before moving.

- One small tray of materials per team. No shopping is needed for version one.
- Teacher-led. No new student account, grade, or online assignment.
- Keep student names off public screens. Use class storage codes.

Previous: 9. Tank 3: Try three ideas

Next: 11. Tank 5: Practice the tricky part

NOTES FOR NEXT TIME

Tank 5: Practice the tricky part

I can practice one connection or moving part before using it in my prototype.

Core route lesson tank-joins

MATERIALS

Clean cardboard ■ Paper ■ Pencils ■ Rulers ■ Student scissors ■ Masking tape ■ Reusable trays ■ Craft sticks ■ String ■ Paper fasteners ■ Rubber bands

DEMO IN ONE MINUTE

- Watch a tab, hinge, slider, or string pull.
- Identify where friction, looseness, or weakness could occur.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can practice one connection or moving part before using it in my prototype.
5	Watch and try	Watch a tab, hinge, slider, or string pull. Identify where friction, looseness, or weakness could occur.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Choose the hardest working part in your plan. Build a small practice sample using scrap. Try the movement or connection several times.	Pause each group early. Ask what its next useful action will be.
5	Check one thing	Name the first weak or unreliable point. Change one feature. Keep the stronger sample as a reference.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Demonstrate your practice part and name the improvement.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK

Demonstrate your practice part and name the improvement.

NOTES FOR NEXT TIME

GRADE SUPPORTS

Grade 6 CIRC Tank Compare two mechanisms against the same job and select one using test evidence.

Cleanup. Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Fast finish. Improve the same work or rehearse your explanation. Ask before taking more materials.

Keep moving parts at table level. No launching or pinching tests. Use only teacher-approved tools.

- Next: 12. Tank 6: Build version one**

Tank 6: Build version one

I can build a rough model that shows my idea's main job.

Core route lesson tank-prototype

MATERIALS

Clean cardboard ■ Paper ■ Pencils ■ Rulers ■ Student scissors ■ Masking tape ■ Reusable trays ■ Craft sticks ■ String ■ Paper fasteners

DEMO IN ONE MINUTE

- Watch the main working part being built first.
- Notice that a rough prototype can answer a useful question.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can build a rough model that shows my idea's main job.
5	Watch and try	Watch the main working part being built first. Notice that a rough prototype can answer a useful question.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Build the essential working part from your plan. Add only the support needed to try it. Keep a sketch or picture of version one using the approved classroom method.	Pause each group early. Ask what its next useful action will be. A failed first attempt is useful evidence. Do not add decoration before the working part is testable.
5	Check one thing	Ask a partner to identify the model's job. Try the intended action once. Mark one feature that needs a test next time.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Show version one doing its main job, or identify the exact part that needs repair.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK **Show version one doing its main job, or identify the exact part that needs repair.**

NOTES FOR NEXT TIME

Grade supports, reset, and the conditions that stay visible.

GRADE SUPPORTS

Grade 5 CIRC Tank Jr Build one working feature with teacher support.

Grade 6 CIRC Tank Build a working feature within the agreed size and material limits.

CLEANUP, EXIT, AND EARLY FINISHERS

Cleanup. Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Exit check. Show version one doing its main job, or identify the exact part that needs repair.

Fast finish. Improve the same work or rehearse your explanation. Ask before taking more materials.

SAFETY AND CONDITIONS

Use teacher-approved materials and tools. Keep paths clear. Return tools before moving.

- A failed first attempt is useful evidence. Working part before decoration.
- Teacher-led. No new student account, grade, or online assignment.
- Keep student names off public screens. Use class storage codes.

Previous: 11. Tank 5: Practice the tricky part

Next: 13. Tank 7: Plan a fair test

NOTES FOR NEXT TIME

Tank 7: Plan a fair test

I can plan a repeatable test that checks what my design claims.

Core route lesson tank-test-plan

MATERIALS

Prototype ■ Paper ■ Pencils ■ Ruler or timer if needed ■ Reusable test objects

DEMO IN ONE MINUTE

- Compare two tests that change several things.
- Watch a fair test keep the object, starting point, and action consistent.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can plan a repeatable test that checks what my design claims.
5	Watch and try	Compare two tests that change several things. Watch a fair test keep the object, starting point, and action consistent.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Write or draw what the product claims to do. Choose one thing to count, measure, or observe. Plan three trials with the same starting conditions.	Pause each group early. Ask what its next useful action will be.
5	Check one thing	Walk through the test without rushing. Ask a partner what must stay the same. Get the teacher's approval before the real trials.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Explain your test, what stays the same, and what result would count as success.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK

Explain your test, what stays the same, and what result would count as success.

NOTES FOR NEXT TIME

Grade supports, reset, and the conditions that stay visible.

GRADE SUPPORTS

Grade 5 CIRC Tank Jr Use a teacher-provided test pattern and a simple works/not-yet record.

Grade 6 CIRC Tank Choose a numerical measure and a success threshold before testing.

CLEANUP, EXIT, AND EARLY FINISHERS

Cleanup. Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Exit check. Explain your test, what stays the same, and what result would count as success.

Fast finish. Improve the same work or rehearse your explanation. Ask before taking more materials.

SAFETY AND CONDITIONS

Only conduct teacher-approved tests at table level. No people, valuables, or live animals are test objects.

- Teacher-led. No new student account, grade, or online assignment.
- Keep student names off public screens. Use class storage codes.

Previous: 12. Tank 6: Build version one

Next: 14. Tank 8: Test and record

NOTES FOR NEXT TIME

Tank 8: Test and record

I can run the same test three times and record what actually happens.

Core route lesson tank-testing

MATERIALS

Prototype ■ Approved test
objects ■ Paper ■ Pencils ■ Ruler or timer if needed

DEMO IN ONE MINUTE

■ Watch one trial from the same starting position.
■ See a record include an unsuccessful result honestly.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can run the same test three times and record what actually happens.
5	Watch and try	Watch one trial from the same starting position. See a record include an unsuccessful result honestly.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Reset the test to the agreed starting position. Run three trials and record each result. Keep the design unchanged until the trials are finished.	Pause each group early. Ask what its next useful action will be.
5	Check one thing	Compare all three results. Point to the most useful pattern or failure. Choose one problem to fix next.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Show three results and identify one specific design problem.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK **Show three results and identify one specific design problem.**

NOTES FOR NEXT TIME

Grade supports, reset, and the conditions that stay visible.

GRADE SUPPORTS

Grade 5 CIRC Tank Jr Use pictures, tally marks, or works/not-yet symbols for three trials.

Grade 6 CIRC Tank Use measured results and explain variation between the trials.

CLEANUP, EXIT, AND EARLY FINISHERS

Cleanup. Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Exit check. Show three results and identify one specific design problem.

Fast finish. Improve the same work or rehearse your explanation. Ask before taking more materials.

SAFETY AND CONDITIONS

Use the approved test only. Stop if a part breaks sharply or the setup becomes unstable.

- Teacher-led. No new student account, grade, or online assignment.
- Keep student names off public screens. Use class storage codes.

Previous: 13. Tank 7: Plan a fair test

Next: 15. Tank 9: Improve one feature

NOTES FOR NEXT TIME

Tank 9: Improve one feature

I can use test evidence to change one feature and check whether it helps.

Core route lesson tank-revision

MATERIALS

Clean cardboard ■ Paper ■ Pencils ■ Rulers ■ Student scissors ■ Masking tape ■ Reusable trays ■ Prototype ■ Previous test record ■ Approved test objects

DEMO IN ONE MINUTE

- Compare a design change with a decorative change.
- Trace one failed test to the part causing it.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can use test evidence to change one feature and check whether it helps.
5	Watch and try	Compare a design change with a decorative change. Trace one failed test to the part causing it.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Choose one problem from your test record. Change the relevant shape, connection, or movement. Repeat the same three-trial test.	Pause each group early. Ask what its next useful action will be.
5	Check one thing	Compare the old and new results. Keep the improvement or explain why you will undo it. Record what the evidence suggests next.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Show one evidence-based change and compare its before-and-after results.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK

Show one evidence-based change and compare its before-and-after results.

NOTES FOR NEXT TIME

Tank 10: Let someone try

I can watch a user try my idea and learn without coaching every move.

Core route lesson tank-user-test

MATERIALS

Prototype ■ Paper ■ Pencils ■ Reusable feedback cards ■ Scrap materials

DEMO IN ONE MINUTE

- Watch a user receive one short instruction.
- Notice how the designer listens without taking over.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can watch a user try my idea and learn without coaching every move.
5	Watch and try	Watch a user receive one short instruction. Notice how the designer listens without taking over.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Tell an approved partner the product's job. Let the partner try it using the approved test. Record one clear action, one confusion, and one useful comment.	Pause each group early. Ask what its next useful action will be.
5	Check one thing	Ask what was easiest and hardest. Choose one change that would help the user. Try a quick paper or scrap version of that change.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Explain one user difficulty and the change you will make.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK Explain one user difficulty and the change you will make.

NOTES FOR NEXT TIME

Grade 5 CIRC Tank Jr Use two teacher-provided questions and one partner tryout.

CLEANUP, EXIT, AND EARLY FINISHERS

Exit check. Explain one user difficulty and the change you will make.

SAFETY AND CONDITIONS

- Teacher-led. No new student account, grade, or online assignment.
- Keep student names off public screens. Use class storage codes.

Next: 17. Tank 11: Use materials wisely

Tank 11: Use materials wisely

I can choose materials that do the job while reducing unnecessary waste.

Core route lesson tank-materials

MATERIALS

Clean cardboard ■ Paper ■ Pencils ■ Rulers ■ Student scissors ■ Masking tape ■ Reusable trays ■ Safe material samples ■ Prototype ■ Shared scale if available

DEMO IN ONE MINUTE

- Compare two materials doing the same small job.
- Notice what can be reused or made smaller.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can choose materials that do the job while reducing unnecessary waste.
5	Watch and try	Compare two materials doing the same small job. Notice what can be reused or made smaller.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Sort the design's parts into essential and optional. Compare two suitable materials for one part. Improve that part without making the product harder to use.	Pause each group early. Ask what its next useful action will be. Use qualitative comparisons if no scale exists. Do not invent purchase costs or environmental claims.
5	Check one thing	Check that the main job still works. Count pieces or compare material size before and after. Sort useful scraps for another group.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Show a useful material choice and explain what you saved or improved.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK **Show a useful material choice and explain what you saved or improved.**

NOTES FOR NEXT TIME

GRADE SUPPORTS

Grade 6 CIRC Tank Explain the tradeoff among performance, quantity, reuse, and finish.

Fast finish. Improve the same work or rehearse your explanation. Ask before taking more materials.

- Use qualitative comparisons if no scale exists. Do not invent costs or environmental claims.
- Teacher-led. No new student account, grade, or online assignment.
- Keep student names off public screens. Use class storage codes.

Next: 18. Tank 12: Name and package

Tank 12: Name and package

I can communicate my product's purpose with a clear name, logo, and simple package.

Core route lesson tank-brand-package

MATERIALS

Prototype ■ Paper ■ Shared markers ■ Reclaimed boxes ■ Cardboard ■ Tape ■ Scissors ■ Ruler

DEMO IN ONE MINUTE

- Compare a readable product label with a crowded one.
- Watch a package protect the model and open easily.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can communicate my product's purpose with a clear name, logo, and simple package.
5	Watch and try	Compare a readable product label with a crowded one. Watch a package protect the model and open easily.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Choose a clear product name and reuse your logo skills. Make a label that tells the product's job. Create or improve a simple protective package.	Pause each group early. Ask what its next useful action will be.
5	Check one thing	Ask a partner to read the label without help. Check opening, fit, and the approved handling test. Revise one confusing or wasteful detail.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Show a readable product label and a package that opens and protects.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK **Show a readable product label and a package that opens and protects.**

NOTES FOR NEXT TIME

Grade supports, reset, and the conditions that stay visible.

GRADE SUPPORTS

Grade 5 CIRC Tank Jr Use one name, one symbol, and one short purpose statement.

Grade 6 CIRC Tank Connect branding and packaging choices to the user and test evidence.

CLEANUP, EXIT, AND EARLY FINISHERS

Cleanup. Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Exit check. Show a readable product label and a package that opens and protects.

Fast finish. Improve the same work or rehearse your explanation. Ask before taking more materials.

SAFETY AND CONDITIONS

Use the approved gentle handling test. No throwing or heated forming without a separate approved teacher station.

- Teacher-led. No new student account, grade, or online assignment.
- Keep student names off public screens. Use class storage codes.

Previous: 17. Tank 11: Use materials wisely

Next: 19. Tank 13: Explain your idea

NOTES FOR NEXT TIME

Tank 13: Explain your idea

I can explain the problem, my solution, and evidence that helped me improve it.

Core route lesson tank-pitch

MATERIALS

Prototype ■ Previous test record ■ Index cards ■ Pencils ■ Product label

DEMO IN ONE MINUTE

- Hear a short pitch with a problem, idea, proof, and next step.
- Notice a speaker showing the product while explaining it.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can explain the problem, my solution, and evidence that helped me improve it.
5	Watch and try	Hear a short pitch with a problem, idea, proof, and next step. Notice a speaker showing the product while explaining it.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Make four short cue cards or picture prompts. Choose one clear test result as proof. Prepare a simple demonstration of the product.	Pause each group early. Ask what its next useful action will be. No sales claims or money request is needed. A classroom pitch explains design decisions.
5	Check one thing	Tell the story to one partner. Ask the partner to repeat the main idea. Cut details that hide the problem, solution, or proof.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Give a short practice pitch that includes the problem, solution, and one piece of evidence.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK Give a short practice pitch that includes the problem, solution, and one piece of evidence.

NOTES FOR NEXT TIME

Grade supports, reset, and the conditions that stay visible.

GRADE SUPPORTS

Grade 5 CIRC Tank Jr Use three picture prompts for a 30-45 second supported explanation.

Grade 6 CIRC Tank Use a roughly 60-second explanation with evidence, a design tradeoff, and a realistic next step.

CLEANUP, EXIT, AND EARLY FINISHERS

Cleanup. Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Exit check. Give a short practice pitch that includes the problem, solution, and one piece of evidence.

Fast finish. Improve the same work or rehearse your explanation. Ask before taking more materials.

SAFETY AND CONDITIONS

Use teacher-approved materials and tools. Keep paths clear. Return tools before moving.

- A classroom pitch explains design decisions. No sales claim and no money request.
- Teacher-led. No new student account, grade, or online assignment.
- Keep student names off public screens. Use class storage codes.

Previous: 18. Tank 12: Name and package

Next: 20. Tank 14: Rehearse and refine

NOTES FOR NEXT TIME

Tank 14: Rehearse and refine

I can practice a clear presentation and improve it after specific feedback.

Core route lesson tank-rehearse

MATERIALS

Prototype ■ Cue cards ■ Existing classroom timer ■ Reusable feedback cards ■ Display surface

DEMO IN ONE MINUTE

■ Watch a calm handoff and a table-level demonstration.
■ Compare useful feedback with a vague compliment.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can practice a clear presentation and improve it after specific feedback.
5	Watch and try	Watch a calm handoff and a table-level demonstration. Compare useful feedback with a vague compliment.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Choose who will explain or demonstrate each part. Rehearse once while a partner times the explanation. Practice one likely question and an honest answer.	Pause each group early. Ask what its next useful action will be. Students may point, demonstrate, or use a partner. Speaking alone is not the only way to participate.
5	Check one thing	Ask for one strength and one specific improvement. Change the unclear part. Rehearse again with a reliable reset.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Deliver your revised pitch and show how the product resets for the next visitor.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK

Deliver your revised pitch and show how the product resets for the next visitor.

NOTES FOR NEXT TIME

44

Tank 15: Share and reflect

I can show my design, listen to feedback, and explain how it changed.

Core route lesson tank-showcase

MATERIALS

Prototype ■ First sketch ■ Test evidence ■ Labels ■ Existing display surfaces ■ Paper ■ Pencils

DEMO IN ONE MINUTE

- Watch a visitor approach, hear a short pitch, and ask one question.
- Notice how the presenter resets the space.

Min	Step	What students actually do	Teacher move
3	Get ready	Read today's goal. Look at the example. Tell your partner one useful detail.	Say today's goal in plain words: I can show my design, listen to feedback, and explain how it changed.
5	Watch and try	Watch a visitor approach, hear a short pitch, and ask one question. Notice how the presenter resets the space.	Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.
17	Make it work	Set out the prototype, label, and one piece of test evidence. Take turns presenting and visiting other teams. Give one useful observation and one respectful question.	Pause each group early. Ask what its next useful action will be. This is an in-class share unless a separate audience or event has been approved. Do not promise outside visitors.
5	Check one thing	Compare your first sketch with your final version. Choose the most useful change. Name one thing you would test next.	Ask for the evidence named in this check. Give one specific next step.
3	Reset together	Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.	Call tool returns first. Check table trays and keep unfinished work for the next contact.
2	Show your result	Explain how evidence changed your design and identify a sensible next test.	Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

EXIT CHECK Explain how evidence changed your design and identify a sensible next test.

NOTES FOR NEXT TIME

Grade supports, reset, and the conditions that stay visible.

GRADE SUPPORTS

Grade 5 CIRC Tank Jr Show the first and final versions and describe one improvement.

Grade 6 CIRC Tank Explain a supported improvement, remaining limitation, and next test.

CLEANUP, EXIT, AND EARLY FINISHERS

Cleanup. Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Exit check. Explain how evidence changed your design and identify a sensible next test.

Fast finish. Improve the same work or rehearse your explanation. Ask before taking more materials.

SAFETY AND CONDITIONS

Keep demonstrations at the approved stations. Reset loose parts and walk between displays.

- In-class share unless a separate audience has been approved. Do not promise outside visitors.
- Teacher-led. No new student account, grade, or online assignment.
- Keep student names off public screens. Use class storage codes.

Previous: 20. Tank 14: Rehearse and refine

Next: 22. Outdoor Microclimate Map

NOTES FOR NEXT TIME

Outdoor Microclimate Map

Observe and compare outdoor places. Thermometers are optional.

Catalog project 6: Outdoor Microclimate Map (existing timed plan)

MATERIALS

Clipboards ■ Outdoor zone map ■ Pencils ■ Thermometers if available ■ Wind ribbons ■ Observation key ■ Indoor weather data cards

DEMO IN ONE MINUTE

"One outdoor classroom can contain several tiny climates at the same time."

- Mark the approved outdoor zones and model one sun, shade, wind, surface, and comfort observation.
- Assign each team two zones and demonstrate safe thermometer or wind-ribbon use when those tools are available.
- Provide indoor data cards if weather or air quality prevents outdoor work.

Min	Step	What students actually do	Teacher move
3	Get Ready	Clip the zone map to your board. Check your assigned zones.	One outdoor classroom can contain several tiny climates at the same time. A useful map combines measurement, observation, and a recommendation for a real activity. Provide indoor data cards if weather or air quality prevents outdoor work.
2	Walk Out	Walk with your weather team.	—
3	Use Tools Safely	Carry the thermometer by its case. Leave unknown plants untouched. Stay inside each marked zone.	Mark the approved outdoor zones and model one sun, shade, wind, surface, and comfort observation.
8	Observe Two Zones	Mark sun or shade. Mark the surface condition. Watch the wind ribbon.	Assign each team two zones and demonstrate safe thermometer or wind-ribbon use when those tools are available.
6	Take Readings	Take one approved reading in each zone. Write each number on the map.	—
5	Build The Map	Add both zones to the class key. Choose the best zone for one activity.	—
3	Pack The Tools	Return each tool to its case. Clip all maps together.	—
2	Return Inside	Walk back with your weather team.	—
3	Exit Check	Write two observations that support your zone choice.	—

EXIT CHECK **Point to two mapped zones and defend one activity recommendation with evidence.**

NOTES FOR NEXT TIME

Full objective. Compare conditions across outdoor classroom zones and recommend where different learning activities should happen.

Grade 5 CIRC Tank Jr Assign the two zones and hand over the finished map key. Accept sun or shade, wind and surface marks, plus one activity choice backed by two circled clues.

Outdoor Microclimate Map: Indoor weather-card version

Use when weather or air quality keeps the class inside.

Catalog project 6: Outdoor Microclimate Map (existing timed plan) | Complete 35-minute lesson. Same grade supports and conditions as the page before.

MATERIALS

Clipboards ■ Outdoor zone map ■ Pencils ■ Thermometers if available ■ Wind ribbons ■ Observation key ■ Indoor weather data cards

DEMO IN ONE MINUTE

"One outdoor classroom can contain several tiny climates at the same time."

- Mark the approved outdoor zones and model one sun, shade, wind, surface, and comfort observation.
- Assign each team two zones and demonstrate safe thermometer or wind-ribbon use when those tools are available.
- Provide indoor data cards if weather or air quality prevents outdoor work.

Min	Step	What students actually do	Teacher move
3	Get Ready	Place the indoor map on your table. Pick up two weather data cards.	Say why the class is staying inside today.
4	Keep Cards Safe	Keep every card on the table. Use no outdoor tool. Walk only when the teacher says go.	Show where cards and pencils stay.
5	Learn The Map Key	Find the sun mark. Find the shade mark. Find the wind mark.	Model one card with the class map key.
7	Compare Two Zones	Read the first weather data card. Mark its sun, wind, and ground clues. Do the same with the second card.	Read one sample card aloud.
7	Build The Map	Add both zones to the indoor map. Copy each number carefully. Check the map with a partner.	Check one team before all teams continue.
4	Choose A Zone	Pick the best zone for one activity. Circle two clues that support your choice.	Ask teams to use two clues, not a guess.
2	Reset The Table	Stack the weather data cards. Return the map and pencil.	Count the card sets.
3	Exit Check	Name the zone you picked. Read your two clues.	Listen for a choice and two matching clues.

EXIT CHECK

Point to two mapped zones and defend one activity recommendation with evidence.

NOTES FOR NEXT TIME

Seed Travelers

Test a seed-inspired carrier with shared test stations.

Catalog project 7: Seed Travelers (existing timed plan)

MATERIALS

Teacher-provided seed samples or image cards ■ Paper ■ Coffee filters ■ Cotton ■ Yarn ■ Paper clips ■ Tape ■ Fan ■ Water tray ■ Fabric test strip

DEMO IN ONE MINUTE

"Plants cannot walk, but their seeds use shape, air, water, animals, and gravity to move."

- Show teacher-provided seed samples or image cards without collecting from living plants.
- Model matching one seed feature to one travel strategy.
- Set wind, drop, float-tray, and fabric hitchhiker test stations with clear boundaries.

Min	Step	What students actually do	Teacher move
3	Get Ready	Choose one seed sample card. Place the test sheet on your table.	Plants cannot walk, but their seeds use shape, air, water, animals, and gravity to move. A model is successful when its structure matches the travel strategy being tested.
3	Test Safely	Keep water inside the tray. Wipe each spill with a towel. Keep every sample away from your mouth.	Show teacher-provided seed samples or image cards without collecting from living plants. Set wind, drop, float-tray, and fabric hitchhiker test stations with clear boundaries.
4	Sort The Seeds	Study the seed shape. Choose wind, drop, float, or hitchhiking. Record your reason.	Model matching one seed feature to one travel strategy.
9	Build A Model	Sketch one model seed. Choose materials that match the travel method. Build the model.	—
7	Run Three Trials	Use the matching test station. Run one trial at a time. Record each result.	—
4	Change One Part	Change one shape or material. Repeat the same test.	—
2	Reset The Station	Dry the wet station. Return every model piece.	—
3	Exit Check	Circle the better result. Name the useful feature.	—

EXIT CHECK Name the travel strategy, show the trial evidence, and explain how one model feature supported movement.

NOTES FOR NEXT TIME

Full objective. Investigate seed-dispersal strategies and build a model seed that travels by wind, drop, float, or hitchhiking simulation.

Grade 5 CIRC Tank Jr	Narrow the choice to two travel strategies. Accept three tally results and one named feature that helped the model move.
Grade 6 CIRC Tank	Require the same station for all trials, a written result each time, and an explanation of why the changed feature did or did not help.

Cleanup. Dry the float station, return samples and image cards, collect model pieces, and sort reusable materials.

Exit check. Name the travel strategy, show the trial evidence, and explain how one model feature supported movement.

Fast finish. New Habitat: Draw the habitat where the model seed would have the best chance to travel and begin growing.

Stretch. Compare two dispersal strategies and argue which one works better under a stated environmental condition.

Use only teacher-provided samples, do not taste any material, keep water at the wet station, and release models only into the marked test area.

- Use teacher-provided seed samples or image cards. Nothing is collected from living plants.
- Water stays at the float station. Nothing goes in a mouth.

Next: 24. Paper Bridge

Paper Bridge

Compare a small structure using the same reusable loads.

Catalog project 5: Paper Bridge (existing timed plan)

MATERIALS

Copy paper ■ Index cards ■ Tape ■ Two support blocks ■ Uniform load pieces ■ Recording sheet

DEMO IN ONE MINUTE

"The material stays the same, but its shape can completely change its strength."

- Compare a flat strip, folded beam, and rolled beam across the same gap.
- Model adding load one unit at a time at the center.
- Stop tests before falling weights create a hazard.

Min	Step	What students actually do	Teacher move
3	Get Ready	Place the support blocks on the marks. Count the load pieces.	The material stays the same, but its shape can completely change its strength. A fair test changes the load while keeping the span and setup the same.
3	Test Safely	Keep hands below the table edge. Place only approved loads on the bridge.	Compare a flat strip, folded beam, and rolled beam across the same gap.
5	Plan The Span	Choose a beam shape. Predict where the paper may bend. Sketch the bridge.	Model adding load one unit at a time at the center.
11	Build The Bridge	Fold the paper into your chosen shape. Tape only the planned joints. Place the bridge across the fixed span.	Stop tests before falling weights create a hazard.
7	Load The Bridge	Add one load piece. Record the supported total. Repeat until the bridge fails. Change one weak part.	—
3	Reset The Test	Return every load piece. Recycle damaged paper.	—
3	Exit Check	Write the highest supported load. Name your useful change.	—

EXIT CHECK

Report the maximum load and explain how one shape helped carry it.

NOTES FOR NEXT TIME

Full objective. Transform paper into a bridge that spans a gap and carries increasing loads.

Grade 5 CIRC Tank Jr Offer the beam shape as a choice of three. Record the load count and name the shape that carried it.

CLEANUP, EXIT, AND EARLY FINISHERS

Stretch. Graph load against deflection and identify the point where the bridge begins to fail.

- Stop a test before a falling load becomes a hazard.

Next: 25. Moving Picture Machine

Moving Picture Machine

Make and revise a short paper animation.

Catalog project 18: Moving Picture Machine (existing timed plan)

MATERIALS

Index cards ■ Paper strips ■ Cardstock
circles ■ Pencils ■ Markers ■ Brass fasteners ■ Binder
clips

DEMO IN ONE MINUTE

"Motion can emerge when the brain sees a rapid sequence of slightly different images."

- Model a two-frame flip image and a multi-frame sequence.
- Show registration marks that keep frames aligned.
- Demonstrate a safe spinning or flipping motion.

Min	Step	What students actually do	Teacher move
3	Get Ready	Choose flip, wheel, or strip motion. Place the frame paper in order.	Motion can emerge when the brain sees a rapid sequence of slightly different images. Consistent alignment makes each change easier to see.
3	Fastener Safety	Keep the device at table level. Use fasteners only through marked holes. Ask the teacher about a sharp fastener.	Model a two-frame flip image and a multi-frame sequence.
4	Plan The Motion	Draw a clear start pose. Draw a clear finish pose. Mark the matching dot.	Show registration marks that keep frames aligned.
9	Draw The Frames	Change one small part per frame. Keep each drawing on the registration mark. Number the frames.	Demonstrate a safe spinning or flipping motion.
7	Assemble The Motion	Stack or attach the frames in order. Secure the approved fastener. Move the device slowly.	—
4	Smooth The Motion	Find the largest visual jump. Revise one frame. Test at two speeds.	—
2	Sort The Pieces	Return unused fasteners. Clip your frames together.	—
3	Exit Check	Show the finished motion once. Name the frame you changed.	—

EXIT CHECK Run the animation and identify one frame changed to make the motion smoother.

NOTES FOR NEXT TIME

Grade supports, reset, and the conditions that stay visible.

Full objective. Create a sequence device that uses small image changes to produce the illusion of motion.

GRADE SUPPORTS

Grade 5 CIRC Tank Jr Use four to six frames on teacher-marked registration dots. One small change per frame.

Grade 6 CIRC Tank Use more frames, test two speeds, and explain which frame change smoothed the motion.

CLEANUP, EXIT, AND EARLY FINISHERS

Cleanup. Cap markers, collect fasteners, and store finished sequences flat.

Exit check. Run the animation and identify one frame changed to make the motion smoother.

Fast finish. Smooth Loop: Add transition frames so the final image flows naturally back to the first.

Stretch. Use timing marks to compare how playback speed changes the perceived motion.

SAFETY AND CONDITIONS

Keep spinning devices at table level and handle fasteners only as demonstrated.

- Spinning and flipping stay at table level.
- Fasteners go through the marked holes only. A sharp fastener goes to the teacher.

Previous: 24. Paper Bridge

Next: 26. Hydroponics Flow Lab

NOTES FOR NEXT TIME

Hydroponics Flow Lab

Test a gravity-fed water model, not a purchased growing installation.

Catalog project 26: Hydroponics Flow Lab (existing timed plan)

MATERIALS

Clean tubing ■ Cups ■ Reclaimed channels ■ Water trays ■ Measuring cups ■ Towels ■ Plant-site models ■ Teacher-controlled tubing cutter ■ Wet mats ■ Sink bucket

DEMO IN ONE MINUTE

"Plants need access to water and air, so more water is not automatically better."

- Model a dry layout before adding any water.
- Demonstrate a measured pour and leak check at every connection.
- Keep all water testing inside the protected wet zone.

Min	Step	What students actually do	Teacher move
3	Get Ready	Place the dry tray on the wet mat. Measure the test water.	Plants need access to water and air, so more water is not automatically better. A good flow model makes distribution visible and measurable.
4	Wet Tool Safety	Keep water inside the tray. Wipe each spill with a towel. Do not cut tubing yourself. Keep electronics away from the wet zone.	Model a dry layout before adding any water.
3	Plan The Flow	Draw the water cup. Draw the water path. Mark each plant spot and the end cup.	Demonstrate a measured pour and leak check at every connection.
7	Build The Dry Model	Mark the cut line on the tube. Give the tube to the teacher. Wait for the cut, then connect the tube path. Check every connection while dry.	Keep all water testing inside the protected wet zone.
6	Run The Water	Pour the measured water into the water cup. Watch each plant spot. Measure the collected water.	—
6	Change The Flow	Change one slope, hole, or loose connection. Repeat with the same water amount. Record the new collected amount.	—
4	Dry The Lab	Drain the tray into the sink bucket. Place wet parts on the drying tray. Dry the table and floor.	—
2	Exit Check	Write the collected amount. Name the flow change.	—

EXIT CHECK

Show the measured flow result and explain the one change that made distribution more even.

NOTES FOR NEXT TIME

Grade supports, reset, and the conditions that stay visible.

Full objective. Model a gravity-fed hydroponic path that distributes water evenly without leaking or flooding.

GRADE SUPPORTS

Grade 5 CIRC Tank Jr Hand over the dry layout template. Compare the collected water as more or less than before.

Grade 6 CIRC Tank Record both measured amounts, name the single change, and state one limit of the model.

CLEANUP, EXIT, AND EARLY FINISHERS

Cleanup. Drain models into the assigned container, dry all surfaces, and separate wet materials for drying.

Exit check. Show the measured flow result and explain the one change that made distribution more even.

Fast finish. Flow Alarm: Design a visible marker that shows when one plant site receives too much or too little water.

Stretch. Calculate the percent of starting water recovered and identify where losses occur.

SAFETY AND CONDITIONS

Keep water away from outlets and electronics, wipe spills immediately, and use no pump unless the teacher controls it.

- This is a gravity-fed water model, not a purchased growing installation.
- The teacher makes every tubing cut. Students mark the line and hand the tube over.
- No pump unless the teacher controls it. Water stays away from outlets and electronics.

Previous: 25. Moving Picture Machine

Next: 27. Stormwater Rescue

NOTES FOR NEXT TIME

Stormwater Rescue

Use shared trays to compare runoff before and after one change.

Catalog project 27: Stormwater Rescue (existing timed plan)

MATERIALS

Sloped trays ■ Model soil or sand ■ Measured water cups ■ Collection cups ■ Gravel ■ Sponge ■ Craft sticks ■ Mesh ■ Model plants ■ Towels ■ Scissors ■ Wet mats ■ Return tub

DEMO IN ONE MINUTE

"Rain becomes a design problem when water moves too quickly across a surface and carries soil with it."

- Set identical sloped trays and model the baseline rain test with a fixed water volume and pour time.
- Show how to measure runoff volume, visible soil movement, or time to collection.
- Demonstrate one barrier, absorbent zone, channel, or plant-root model without presenting a single correct answer.

Min	Step	What students actually do	Teacher move
3	Get Ready	Place the sloped tray on the wet mat. Measure the rain water.	Rain becomes a design problem when water moves too quickly across a surface and carries soil with it. The goal is not drinkable water. The goal is evidence that one intervention changes runoff or erosion in the model.
4	Stormwater Safety	Keep wet material inside the tray. Wipe each spill with a towel. Cut mesh only on the table. Keep fingers outside the cutting path. Taste no test material.	Set identical sloped trays and model the baseline rain test with a fixed water volume and pour time.
5	Run The First Test	Pour the measured rain from the marked height. Measure the water that leaves the tray. Mark where soil moved.	Show how to measure runoff volume, visible soil movement, or time to collection.
6	Build A Rescue	Choose one problem location. Cut only the planned mesh piece. Build one helper that slows or moves the water.	Demonstrate one barrier, absorbent zone, channel, or plant-root model without presenting a single correct answer.
6	Repeat The Rain	Pour the same water amount. Measure the water that leaves the tray. Mark where soil moved this time.	—
3	Improve The Rescue	Change one part of the helper. Predict the next result.	—
5	Clean The Tray	One table at a time, carry wet soil to the return tub. Place wet tools on the drying mat. Dry the tray and table. Wash hands when the teacher calls your table.	—
3	Exit Check	Compare both water amounts. Name the helpful feature.	—

EXIT CHECK

Compare baseline and revised evidence, then state what the intervention changed and what the model cannot prove.

Grade supports, reset, and the conditions that stay visible.

Full objective. Model stormwater runoff and erosion, then test one intervention that slows, redirects, absorbs, or filters the flow.

GRADE SUPPORTS

Grade 5 CIRC Tank Jr Keep the same tray and offer two pictured intervention options. Compare before and after by what is visible in the tray.

Grade 6 CIRC Tank Record runoff volume both times, state what the intervention changed, and name what the model cannot prove.

CLEANUP, EXIT, AND EARLY FINISHERS

Cleanup. Pour water only into the assigned container, return reusable media, dispose of model soil as directed, and dry the full wet zone.

Exit check. Compare baseline and revised evidence, then state what the intervention changed and what the model cannot prove.

Fast finish. Storm Upgrade: Predict how the intervention would perform with twice the water and identify the first likely failure.

Stretch. Calculate the percent change in runoff volume or collection time and explain the limitation of the model.

SAFETY AND CONDITIONS

Never drink test water, keep wet materials inside trays, wipe spills immediately, and wash hands after handling model soil.

- Six shared trays. Test water is never tasted and never leaves the tray.
- Model soil is returned as directed and hands are washed by table.

Previous: 26. Hydroponics Flow Lab

Next: 28. Micro:bit Sensor Station

NOTES FOR NEXT TIME

Micro:bit Sensor Station

Use existing approved devices or the complete paper-code fallback.

Catalog project 28: Micro:bit Sensor Station (existing timed plan)

MATERIALS

Classroom micro:bits or confirmed equivalents ■ USB cables ■ Computers ■ Teacher-approved sensors if available ■ Starter code cards ■ Sensor data cards ■ Paper code blocks ■ Test record

DEMO IN ONE MINUTE

"A sensor turns a condition in the world into data a program can use."

- Confirm the available micro:bit or equivalent hardware and demonstrate one input, threshold rule, and output.
- Provide starter code blocks for light, temperature, sound, movement, or a teacher-approved external sensor.
- Use device-free sensor data cards and paper code blocks as the fallback if hardware or network access is unavailable.

Min	Step	What students actually do	Teacher move
5	Get Ready	Place the micro:bit beside the computer. Open the teacher starter file. Connect only the approved cable. Wait until the computer shows the board.	A sensor turns a condition in the world into data a program can use. A useful alert has a clear threshold, a clear user, and a response that can be tested. Use device-free sensor data cards and paper code blocks as the fallback if hardware or network access is unavailable.
4	Device Safety	Use only approved USB power. Connect no unknown device. Handle the board by its edges. Disconnect power when the teacher says stop.	Confirm the available micro:bit or equivalent hardware and demonstrate one input, threshold rule, and output.
5	Plan The Sensor	Choose one classroom condition. Write what the board will measure. Choose the number that turns the alert on. Choose a clear alert.	Provide starter code blocks for light, temperature, sound, movement, or a teacher-approved external sensor.
6	Build The Program	Place the measure block first. Add the number check. Add the alert block. Transfer the program to the approved device.	—
6	Test Three Values	Test one number below the turn-on number. Test the turn-on number. Test one number above it. Record each alert.	—
3	Tune The Alert	Change the turn-on number or alert. Repeat the three tests.	—
3	Disconnect Safely	Stop the program. Unplug the USB cable by its connector. Return the board and cable to the tray.	—
3	Exit Check	State the final turn-on number. Name one limit of the data.	—

EXIT CHECK

Demonstrate three input tests and explain the sensor, threshold, output, and one revision.

Full objective. Program a classroom micro:bit or confirmed equivalent to detect a condition and produce a useful visual, sound, or motion alert.

Grade 5 CIRC Tank Jr Provide the starter blocks and one condition. Name the turn-on number and the alert.

Micro:bit Sensor Station: Paper-code version

Use when a computer, cable, or micro:bit is not ready.

Catalog project 28: Micro:bit Sensor Station (existing timed plan) | Complete 35-minute lesson. Same grade supports and conditions as the page before.

MATERIALS

Classroom micro:bits or confirmed equivalents ■ USB cables ■ Computers ■ Teacher-approved sensors if available ■ Starter code cards ■ Sensor data cards ■ Paper code blocks ■ Test record

DEMO IN ONE MINUTE

"A sensor turns a condition in the world into data a program can use."

- Confirm the available micro:bit or equivalent hardware and demonstrate one input, threshold rule, and output.
- Provide starter code blocks for light, temperature, sound, movement, or a teacher-approved external sensor.
- Use device-free sensor data cards and paper code blocks as the fallback if hardware or network access is unavailable.

Min	Step	What students actually do	Teacher move
3	Get Ready	Place the paper code blocks on the mat. Pick up three sensor data cards.	Say: We can learn the same idea without a device.
4	Keep Pieces Safe	Keep every paper block on the mat. Use no cable or device. Pick up dropped cards at once.	Show the paper input, check, and alert blocks.
5	Plan The Alert	Pick one classroom condition. Pick the number that turns the alert on. Draw the alert.	Explain: The turn-on number is called the threshold.
8	Build Paper Code	Place the measure block first. Place the number check next. Place the alert block last. Read the paper code aloud.	Check the block order with each team.
6	Test Three Cards	Read the first sensor data card. Run it through the paper code. Record the alert. Repeat for two more cards.	Model one data card before teams test.
4	Fix The Alert	Change the turn-on number or alert. Test the three cards again.	Ask whether the alert turns on too soon or too late.
2	Reset The Cards	Sort the paper code blocks. Stack the sensor data cards.	Count both card sets.
3	Exit Check	Write the final turn-on number. Name one card that made the alert turn on.	Listen for one number and one matching test card.

EXIT CHECK Demonstrate three input tests and explain the sensor, threshold, output, and one revision.

NOTES FOR NEXT TIME

Cardboard Arcade

Combine the year's connection and testing skills in a compact game.

Catalog project 10: Cardboard Arcade (existing timed plan)

MATERIALS

Cardboard ■ Paper cups ■ Craft sticks ■ Paper fasteners ■ String ■ Tape ■ Markers ■ Marbles or contained game tokens ■ Score cards ■ Tabletop game cards ■ Play trays

DEMO IN ONE MINUTE

"A fun game is a system: the player understands the goal, the mechanism behaves predictably, and the next player can reset it."

- Show compact examples such as a marble maze, spinner target, ring challenge, ramp, or contained token drop.
- Model a start state, player action, scoring rule, and ten-second reset.
- Stop teams after the first playable version for a partner test before decoration.

Min	Step	What students actually do	Teacher move
3	Get Ready	Choose one tabletop game card. Place tokens inside the play tray.	A fun game is a system: the player understands the goal, the mechanism behaves predictably, and the next player can reset it. Start with one playable action, not a giant theme.
3	Game Safety	Keep tokens inside the table boundary. Stop any part that pinches. Use no launcher.	Show compact examples such as a marble maze, spinner target, ring challenge, ramp, or contained token drop.
4	Plan The Game	Write the game goal in one sentence. Sketch the token path.	Model a start state, player action, scoring rule, and ten-second reset.
9	Build The Game	Build the smallest playable version. Add one reliable cardboard connection. Make a clear start point.	Stop teams after the first playable version for a partner test before decoration.
7	Play Test	Ask a partner to play without help. Watch the first confusing moment. Record one unreliable part.	—
4	Improve The Reset	Change the confusing part. Reset the game twice.	—
2	Reset The Arcade	Count every token. Place the game on the storage shelf.	—
3	Exit Check	State the goal. Show how the game resets.	—

EXIT CHECK

Let a partner play, show the ten-second reset, and name the revision that improved the experience.

NOTES FOR NEXT TIME

Full objective. Build a small tabletop cardboard skill game with one clear goal, a reliable mechanism, and a fast reset.

Grade 5 CIRC Tank Jr Give the one-sentence goal frame. Prove one clean reset.

- Every token stays inside the tabletop play area. No launchers.
- Stop any mechanism that pinches or sends a part toward a person.

Next: 30. Tech Terrarium

Tech Terrarium

Make a teacher-approved shared contribution while preserving the existing habitat.

Catalog project 2: Tech Terrarium (existing timed plan)

MATERIALS

Teacher-approved collected rocks ■ Wash tubs ■ Water ■ Towels ■ Terrarium container ■ Prepared base materials ■ Plant-safe decorations ■ Paper ■ Pencils ■ Team cards ■ Materials picture ■ Mode cards ■ Job-area signs ■ Team trays

DEMO IN ONE MINUTE

"Today the class turns ordinary outdoor materials into one shared build."

- Review outdoor collection boundaries and show which rocks and loose natural materials are safe to collect.
- Model washing and drying one rock, then demonstrate the base layers and the empty shared layout.
- Assign collection, wash, base, and layout jobs so every student contributes without crowding the terrarium.

Min	Step	What students actually do	Teacher move
3	Get Ready	Look at the saved terrarium. Touch nothing yet.	Say: Today we keep the old build and add one safe part.
3	Choose Jobs	Read one refresh job card. Go to your team spot.	Give each student one refresh job card.
4	Keep It Safe	Keep all glued parts together. Do not pull or pry. Move only loose parts the teacher marked.	Point to glued parts. Say: These stay together.
5	Look And Record	Find one part that should stay. Find one spot that can work better. Draw what you see.	Show how to look without moving fixed parts.
5	Plan One Add On	Draw one part that can come back off. Keep space for plants and future tools.	Check that each idea can be removed later.
8	Build Off To The Side	Build the new part away from the terrarium. Use only the tools on your card. Wait for the teacher check.	Approve each part before it touches the terrarium.
3	Test The New Part	Test the new part away from the terrarium. Fix one loose or crowded spot.	Lead the class check away from the terrarium.
2	Reset Materials	Return every tool. Store the new part by itself.	Check that the old build still looks the same.
2	Exit Check	Name one part that stayed. Name the new part your team made.	Ask what changed and what must stay next year.

EXIT CHECK

Name one contribution to the shared terrarium and one class decision that made the final layout safer or more useful.

NOTES FOR NEXT TIME

Grade supports, reset, and the conditions that stay visible.

Full objective. Collect, prepare, and assemble the base and shared layout for a whole-class technology terrarium.

GRADE SUPPORTS

Grade 5 CIRC Tank Jr Draw the part that must stay and one removable add-on. Build it off to the side.

Grade 6 CIRC Tank Explain why the add-on is removable, how it protects plant room, and what the class should preserve.

CLEANUP, EXIT, AND EARLY FINISHERS

Cleanup. Return unused outdoor materials where directed, empty wash tubs, dry the floor and tables, fold towels, and leave the terrarium area clear.

Exit check. Name one contribution to the shared terrarium and one class decision that made the final layout safer or more useful.

Fast finish. Designer's Challenge: Sketch one removable cardboard sign, sensor, light, or story feature that could be added later without crowding the terrarium.

Stretch. Add a labeled input-output diagram for a future electronic feature.

SAFETY AND CONDITIONS

Collect only loose teacher-approved materials. Do not disturb plants, habitats, insects, sharp objects, or unknown materials. Walk while carrying rocks and keep water on the wash station.

- Preserve the existing habitat. Nothing glued is pulled, pried, or dismantled.
- Do not move the animal, disturb its habitat, or add water without the teacher.
- New parts are built away from the terrarium and stored separately.

Previous: 29. Cardboard Arcade

Next: 31. Reflect, reset, and catch up

NOTES FOR NEXT TIME

Reflect, reset, and catch up

Use Demo Day reflection and reset, or complete a missed core contact.

Catalog project 36: Demo Day (existing timed plan)

MATERIALS

Showcase exhibits ■ Role cards ■ Safety checklist ■ Visitor prompts ■ Feedback forms ■ Reset supplies

DEMO IN ONE MINUTE

"Today the studio belongs to the teams who built it."

- Confirm safety, traffic flow, and team roles before opening.
- Observe and record evidence without taking over student explanations.
- Lead the final reset and short reflection after the audience leaves.

Min	Step	What students actually do	Teacher move
3	Get Ready	Report to your assigned station. Read your role card.	Today the studio belongs to the teams who built it. Professional means safe, welcoming, honest about evidence, and ready to reset.
3	Station Safety	Operate only approved demonstrations. Keep every walkway clear. Pause any station that cannot reset safely.	Confirm safety, traffic flow, and team roles before opening.
4	Check The Station	Inspect the prototype. Check the visitor boundary. Practice the emergency pause.	Observe and record evidence without taking over student explanations.
9	Host The Demo	Welcome the visitor. State the design need. Demonstrate the approved action. Show the evidence.	Lead the final reset and short reflection after the audience leaves.
7	Rotate The Roles	Reset after each visitor. Move to the next role card. Answer questions with evidence. Name a limit when needed.	—
4	Close The Station	Thank the final visitor. Complete the last safe reset.	—
2	Restore The Room	Return all role cards. Place every exhibit in storage position.	—
3	Exit Check	Name one strong visitor moment. Write one improvement for next time.	—

EXIT CHECK Submit the final reflection with one audience observation, one success, and one next improvement.

NOTES FOR NEXT TIME

Full objective. Independently host, demonstrate, reset, and reflect on a polished project experience for an audience.

GRADE SUPPORTS

Grade 5 CIRC Tank Jr Use three picture prompts: what we made, what worked, what is next.

Grade 6 CIRC Tank Give one evidence-backed success, one honest limit, and one specific next improvement.

CLEANUP, EXIT, AND EARLY FINISHERS

Cleanup. Complete the full station checklist, inventory all shared materials, and leave the studio ready for its next use.

Exit check. Submit the final reflection with one audience observation, one success, and one next improvement.

Fast finish. Next-Year Note: Record one specific lesson that would help a future studio team start stronger.

Stretch. Synthesize visitor feedback into one evidence-backed recommendation for the next iteration.

SAFETY AND CONDITIONS

Operate only approved demonstrations, keep exits and walkways clear, and pause any station that cannot reset safely.

- This is an in-class share. Visitor means a classmate rotating in from another station.
- An outside audience or event is not approved and is not assumed here.
- Option B: run a missed core contact instead. The reflection still closes the class.

Previous: 30. Tech Terrarium

Last contact in this route. Keep any extra contact as a buffer.

NOTES FOR NEXT TIME

WHAT THIS GUIDE DOES NOT DECIDE

Read this before planning around it

Stated plainly so nothing here gets treated as approved.

- No calendar dates appear here. A contact is one class meeting per class. Closures, assemblies, absences, and makeup days shift dates, not the contact order.
- The 21 prepared lessons are not the whole year on their own. Contacts 22 to 31 reuse ten existing catalog projects, printed here in full.
- All 36 catalog alternatives remain available and unchanged. This route uses ten of them.
- Optional equipment stays optional. No purchase, approval, or installation is assumed.
- The resin unit is dry student planning. A small teacher trial is under consideration only.
- The ChompSaw station needs separate approval and its own operating demonstration.
- Scanner and digitizer hardware is unidentified. Demonstrate the real device.
- Contact 31 is an in-class share or a catch-up contact. No outside audience is approved.
- Do not mark a class complete by date alone. Track the contact number for that class.